

SOP-GEN-3800: Emergency Generator and Fire Pump Testing

Document owner: Utilities Department **Effective date:** 2024-01-30 **Review cycle:** 12 months

Section 1. Purpose and Scope

Governs periodic testing of emergency standby generators and fire pumps to verify readiness.

Section 2. Generator Testing

Clause 2.1 — Emergency generators are load-tested (minimum 30% of nameplate rating) monthly under simulated loss-of-power conditions, with a full annual load-bank test at or near 100% rated load.

Clause 2.2 — A generator failing to start or reach rated voltage/frequency within its design time during a monthly test is a critical finding requiring same-day notification to the Utilities Department and repair before the test cycle is considered complete.

Section 3. Fire Pump Testing

Clause 3.1 — Fire pumps are run-tested weekly (churn/no-flow test) and flow-tested annually against rated capacity and pressure per NFPA 25.

Clause 3.2 — A fire pump failing to reach rated flow/pressure at annual test is reported to the Fire Protection Department immediately; the associated fire protection system's impaired status is posted per SOP-FIRE-610's impairment procedure until repair is verified.

Section 4. Fuel Supply

Clause 4.1 — Diesel generator/fire pump fuel tanks are maintained at a minimum of 66% capacity at all times, with fuel quality (water content, microbial growth) tested annually.

Section 5. Records

Clause 5.1 — Test results are filed to the Utilities File and retained for 3 years, or per NFPA 25 requirements for fire pump records, whichever is longer.